Ajay Venkatesh

+1 516 585 9013 Email LinkedIn GitHub Portfolio

Education

New York University, New York, NY — *Aug 2024 – May 2026*

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

Experience

- **New York University (Novel AI Technologies, Inc.)** — *Aug 2024 – Present*
  Full-Stack AI Software Engineer — New York, US
  - Partnered with research and product stakeholders to build full-stack tools (**React, TypeScript, Python, Node.js, REST APIs**) supporting an NSF-funded life-sciences platform; deployed via **Docker** on **AWS EC2** with **PostgreSQL** data infrastructure.
  - Engineered **Python ETL pipelines** into **PostgreSQL** with concurrency control; processed **250K+** records and built researcher-facing dashboards for monitoring model outputs and experimental telemetry.
  - Integrated **LLM-powered workflows** (**GPT-4 + LangChain**) for document analysis and deployed cloud-hosted **computer vision** models (**YOLOv8, EfficientNet**) for structured biological signal capture from mobile inputs.
  - Shipped production features using **Cursor**, **GitHub Copilot**, and **Claude Code** while validating generated code through reviews, testing, and iterative refinement with cross-functional partners.
- **Amazon** — *May 2025 – Aug 2025*
  Software Development Engineer Intern — Seattle, US
  - Built an **AI agent harness** with **RAG** and an **agentic workflow** that retrieves operational logs to reason over incidents and generate context-aware remediation steps for production systems.
  - Implemented a **Model Context Protocol (MCP)** server for tool-augmented LLM retrieval, improving query efficiency and compute utilization; provisioned **AWS CDK** infrastructure on **ECS Fargate** with **multi-stage CI/CD**.
  - Delivered a **React** and **TypeScript** frontend integrated with backend **REST APIs**; wrote unit and integration tests and participated in design and code reviews with senior engineers.
- **CampusMesh** — *Jan 2026 – Present*
  AI Software Engineer — New York, US
  - Built scalable **TypeScript/Node.js** backend services and **LLM retrieval pipelines (RAG)** for an AI assistant, improving contextual accuracy by **30%** through embedding-based semantic search over structured indexes.
  - Deployed **serverless services** on **AWS** (**Lambda, API Gateway, S3**) across **4 environments**; optimized existing modules to support a growing engineering team through refactoring and documented API contracts.
- **PricewaterhouseCoopers (PwC)** — *Aug 2021 – Aug 2024*
  Senior Software Engineer — Bengaluru, India
  - Designed distributed backend infrastructure on **Microsoft Azure** for a healthcare platform; built an **NLP OCR pipeline** (**LayoutLM**) to automate structured extraction from scientific documents, reducing processing time by **75%**.
  - Mentored engineers on software architecture and testing; optimized parallel ingestion over **100K+** records from **5 hours to 1 hour** for analytics workflows used by operations and research-adjacent teams.

Projects

- **InviLite (AI Agent & Data Platform)** | *Python, AWS Bedrock, MCP, FAISS, Docker, Lambda, Kinesis*
  - Built bespoke tooling for ingestion, classification, and triage workflows; integrated **RAG** (**SVM + FAISS + Bedrock LLM**) into an automated decision pipeline, lifting accuracy from **56% to 92%**.
- **Hospital Management System (Researcher-Facing Full-Stack)** | *React, TypeScript, Node.js, PostgreSQL, AWS SageMaker*
  - Architected a full-stack platform with intuitive UI flows for appointments, lab reports, and patient records; deployed **Hugging Face** model inference on **AWS SageMaker** for real-time diagnostic support.

Skills

- **Programming Languages:** TypeScript, Python, JavaScript (ES6), Java, SQL, C++
- **Full-Stack & Backend:** React, Node.js, Express.js, REST API, Microservices, Full-Stack Systems Architecture
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, Database Infrastructure
- **Cloud & DevOps:** AWS (EC2, Lambda, ECS, S3, CDK, SageMaker, Bedrock), Docker, Vercel, CI/CD, Linux
- **AI/ML & Agents:** LangChain, RAG (FAISS), GPT-4, Hugging Face, PyTorch, LLM Fine-tuning, Model Context Protocol (MCP)
- **AI Coding Tools:** Cursor, GitHub Copilot, Claude Code

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*

An Aquila-optimized SVM Classifier for Diabetes Prediction *(IEEE)*

Certifications

AWS Cloud Practitioner | Stanford Machine Learning (Coursera) | Azure Data Fundamentals